algorithms homework 7

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Question 1. given a binary tree, write an algorithm that finds the maximum sum of a path from the root to the leaves such that no two consecutive nodes (parent and child) can be added to the sum

Answer 1.0.1. we add two variables two each node, `with` and `without`, such that `with` is the route from the node with the maximal sum, and `without` is the route with the maximal sum starting from one of its children (node itself isnt part of the sum)

```
MAX-ROUTE-SUM-MAIN(T):
    MAX-ROUTE-SUM(T)
    return max{T.with, T.without}
MAX-ROUTE-SUM(T):
    if T is a leaf then
        T.with ← T.node
        T.without ← 0
    if T has right then
        MAX-ROUTE-SUM(T.right)
    if T has left then
        MAX-ROUTE-SUM(T.left)
    max_with ← 0
    max_without ← 0
    if T has left then
        max_with ← max{max_with, T.left.without}
        max_without ← max{max_without, T.left.with}
    if T has right then
        max_with ← max{max_with, T.right.without}
        max_without ← max{max_without, T.right.with}
    T.with ← T.node + max_with
    T.without ← T.node + max_without
```

this algorithm has a time complexity of $\Theta(n)$ where n denotes the number of nodes in the tree

Question 2. for the purpose of encoding technical texts of a certain type in the hebrew language, research was done on the frequency of the texts

א	5%
ב	6%
ג	3%
ד	5%
ה	7%
ו	8%
ז	4%
ח	3%
ט	3%
י	6%
כ	3%
ל	4%
מ	4%
נ	4%
ס	4%
ע	3%
פ	5%
צ	4%
ק	5%
ר	6%
ש	3%
ת	5%

you have to provide an optimal binary coding for each letter using huffman coding

Answer 2.0.1.

א	5%	0010
ב	6%	0110
ג	3%	10010
ד	5%	0011
ה	7%	1011
ו	8%	1101
ז	4%	11000
ח	3%	10011
ט	3%	10000
י	6%	0111
כ	3%	10001
ל	4%	11001
מ	4%	11110
נ	4%	11111
ס	4%	11100
ע	3%	01010
פ	5%	0000
צ	4%	11101
ק	5%	0001
ר	6%	1010
ש	3%	01011
ת	5%	0100

Question 3. given an array of normal numbers of size n that represent prices of toys. a kidergarten teacher is interested in buying as many toys as possible with a budget of k

Subquestion 3.1. write an algorithm that solves the problem in $\Theta(n)$ time

```
FIND(A, k):
    s ← ∑A
    if s ≤ k then return A
    a ← SELECT(A, ⌊n/2⌋)
    (A_s, sum_s) ← SUM-SMALL(A, a)  // elements smaller than a
    (A_b, sum_b) ← SUM-BIG(A, a)  // elements bigger than a
    if sum_s > k then return FIND(A_s, k)
    if sum_s = k then return A_s
    return A_s ∪ FIND(A_b, k − sum_s)
```

Subquestion 3.2. explain the algorithm and show its correctness

Answer 3.2.1. the algorithm (almost) halves the array on each call until it finds the subarray of numbers to best add up to k

Subquestion 3.3. analyze the time complexity of the algorithm
the recurrence formula for this algorithm is $T(n) = Cn + T(\frac{n}{2})$ which simplifies to $T(n) \in \Theta(n)$

Question 4. given a matrix of normal numbers of size $n \times k$, picking the cell (i, j) would cost $P_{i,j}$ shekels (P is a given matrix)
given a limited budget (normal number), we wanna pick from each row a number such that the cost (sum) of the chosen cells is maximal

Subquestion 4.1. solve the problem in $\Theta(n \cdot k \cdot budget)$ time using dynamic programming

Answer 4.1.1. we use an additional array B of size $n \times budget$ such that $B_{i,j}$ is defined as the solution to the problem with a budget of j and number of rows i
i couldnt solve this :(

Question 5. given the matrices A_1, A_2, A_3, A_4, A_5 of dimensions $4 \times 5, 5 \times 3, 3 \times 8, 8 \times 3, 3 \times 10$, find the cost of the multiplication of this series of matrices using the optimal method

Answer 5.0.1. running the algorithm DMC from class we get

0	84	297	345	505	
	0	189	297	417	
		0	252	532	the answer is 505
			0	360	
				0	

Question 6. propose a file over $\{a, b, c, d, e, f, g, h\}$ whose optimal prefix code is fixed-length, does a similar file exist for $\{a, b, c, d, e, f, g\}$?